

Final Exam

Macroeconomics II

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Problem 1 (Questions 1–20): Inflation, IS–LM, AD–AS

① Definition. If $p_t \equiv \log P_t$, then inflation is B

- (A) $\pi_t \equiv P_t - P_{t-1}$
- (B) $\pi_t \equiv p_t - p_{t-1}$
- (C) $\pi_t \equiv p_{t+1} - p_t$
- (D) $\pi_t \equiv \log(P_{t-1}/P_t)$

Explanations.

- (A) Wrong: $P_t - P_{t-1}$ depends on units and is not the standard definition.
- (B) Correct: inflation is the change in the log price level.
- (C) Wrong timing: that is next period's inflation.
- (D) Wrong sign: this is negative of inflation.

② Calculation. Price level goes from 250 to 260 in one year. Approx. inflation is A

- (A) 4%
- (B) 10%
- (C) 2.5%
- (D) -4%

Explanations.

- (A) Correct: $(260 - 250)/250 = 0.04$.
- (B) Wrong: would correspond to $25 \rightarrow 27.5$ or similar.
- (C) Wrong: $(260 - 250)/260 \approx 3.85\%$, not 2.5%.

(D) Wrong sign.

③ Fisher relation (ex-ante). The real rate is approximately B

(A) $r_t = i_t + \mathbb{E}_t \pi_{t+1}$

(B) $r_t = i_t - \mathbb{E}_t \pi_{t+1}$

(C) $r_t = \pi_t - i_t$

(D) $r_t = i_t - \pi_t$

Explanations.

(A) Wrong sign: expected inflation lowers the real rate.

(B) Correct: $r \approx i - \mathbb{E}\pi$.

(C) Wrong sign.

(D) Wrong: uses current instead of expected inflation.

④ Calculation. If $i_t = 5\%$ and $\mathbb{E}_t \pi_{t+1} = 3\%$, then $r_t \approx$ B

(A) 8%

(B) 2%

(C) -2%

(D) 3%

Explanations.

(A) Wrong: adds instead of subtracts.

(B) Correct: $5 - 3 = 2\%$.

(C) Wrong sign.

(D) Wrong: equals expected inflation, not real rate.

⑤ Nominal anchor. A nominal anchor is best described as B

(A) a mechanism pinning down real GDP permanently

(B) a mechanism pinning down the price level path (or long-run inflation)

(C) a mechanism pinning down unemployment at zero

(D) a mechanism pinning down money velocity

Explanations.

- (A) Wrong: real GDP is not pinned by a nominal anchor.
- (B) Correct.
- (C) Wrong: natural unemployment is not zero.
- (D) Wrong: velocity can vary.

⑥ Fiscal backing intuition. If expected future primary surpluses fall while nominal debt B is given, then (all else equal) A

- (A) P rises so that B/P falls
- (B) P falls so that B/P rises
- (C) P is unchanged because B is nominal
- (D) inflation must be zero

Explanations.

- (A) Correct: lower backing lowers real value; with nominal B , P rises.
- (B) Wrong sign.
- (C) Wrong: P can adjust to revalue nominal claims.
- (D) Wrong.

⑦ One-time jump vs persistent inflation. A permanent one-time jump in P implies B

- (A) permanently high inflation
- (B) a one-time burst of inflation, then inflation returns to normal
- (C) permanently negative inflation
- (D) inflation is undefined

Explanations.

- (A) Wrong: persistent inflation requires continuing increases.
- (B) Correct.
- (C) Wrong.

(D) Wrong.

⑧ Money demand. In $M/P = L(Y, i)$, the usual signs are B

(A) $L_Y < 0$,
 $L_i > 0$

(B) $L_Y > 0$,
 $L_i < 0$

(C) $L_Y > 0$,
 $L_i > 0$

(D) $L_Y < 0$,
 $L_i < 0$

Explanations.

(A) Wrong: money demand rises with income.

(B) Correct: higher income raises money demand; higher interest lowers it.

(C) Wrong: L_i should be negative.

(D) Wrong.

⑨ LM algebra. If $M/P = kY - hi$, then solving for i gives A

(A) $i = \frac{k}{h}Y - \frac{1}{h}\frac{M}{P}$

(B) $i = \frac{h}{k}Y - \frac{1}{k}\frac{M}{P}$

(C) $i = \frac{1}{h}\frac{M}{P} - \frac{k}{h}Y$

(D) $i = kY - h\frac{M}{P}$

Explanations.

(A) Correct.

(B) Wrong: inverted coefficients.

(C) Wrong sign.

(D) Wrong.

⑩ Numerical LM. Let $M/P = 6$, $k = 1$, $h = 2$, $Y = 8$. Then i equals C

- (A) -1
- (B) 0
- (C) 1
- (D) 2

Explanations.

- (A) Wrong: compute gives $+1$.
- (B) Wrong: would require $Y = M/P = 6$.
- (C) Correct: $i = 0.5 \cdot 8 - 0.5 \cdot 6 = 1$.
- (D) Wrong.

⑪ IS curve sign. In $Y = \bar{A} - ar$ with $a > 0$, higher r implies B

- (A) Y rises
- (B) Y falls
- (C) Y unchanged
- (D) P falls mechanically

Explanations.

- (A) Wrong: higher real rates reduce demand.
- (B) Correct.
- (C) Wrong.
- (D) Wrong: price level is not in IS directly.

⑫ Numerical IS. If $Y = 100 - 20r$ and r rises from 0.02 to 0.03 , then $\Delta Y =$ B

- (A) $+0.2$
- (B) -0.2
- (C) -2
- (D) -0.4

Explanations.

- (A) Wrong sign.
- (B) Correct: $-20 \times 0.01 = -0.2$.
- (C) Wrong magnitude (treating 0.01 as 1).
- (D) Wrong magnitude.

⑬ Deriving AD in IS–LM. If M is fixed, a higher price level P typically leads to B

- (A) higher M/P and higher Y
- (B) lower M/P and lower Y
- (C) lower M/P and higher Y
- (D) no change in Y

Explanations.

- (A) Wrong: M/P falls when P rises.
- (B) Correct: lower real balances raise i and reduce demand.
- (C) Wrong sign for Y .
- (D) Wrong.

⑭ Fiscal expansion in IS–LM (short run). A higher G primarily shifts B

- (A) LM right
- (B) IS right
- (C) LM left
- (D) IS left

Explanations.

- (A) Wrong: money market unchanged.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

⑮ Monetary expansion in IS–LM. Higher M (holding P fixed) shifts A

- (A) LM right/down
- (B) IS right
- (C) LM left/up
- (D) IS left

Explanations.

- (A) Correct.
- (B) Wrong: goods-market autonomous demand unchanged.
- (C) Wrong direction.
- (D) Wrong.

①6 AD–AS diagnosis. A negative supply shock typically produces B

- (A) $Y \uparrow$ and $P \downarrow$
- (B) $Y \downarrow$ and $P \uparrow$
- (C) $Y \uparrow$ and $P \uparrow$
- (D) $Y \downarrow$ and $P \downarrow$

Explanations.

- (A) Wrong.
- (B) Correct (stagflation).
- (C) Wrong: demand shock pattern.
- (D) Wrong: that's contractionary demand shock.

①7 Linear SRAS: if $y = y^{FE} + \alpha(p - p^e)$, then the output gap is A

- (A) $x = y - y^{FE} = \alpha(p - p^e)$
- (B) $x = y + y^{FE}$
- (C) $x = p - p^e$
- (D) $x = \alpha(p^e - p)$

Explanations.

- (A) Correct.
- (B) Wrong.
- (C) Wrong: missing α scaling and units.
- (D) Wrong sign.

⑱ Solve AD–AS once. With AD $y = \bar{y}_D - \phi p$ and SRAS $p = p^e + \lambda(y - y^{FE})$, equilibrium p equals B

- (A) $p = p^e + \lambda(\bar{y}_D - y^{FE})$
- (B) $p = \frac{p^e + \lambda(\bar{y}_D - y^{FE})}{1 + \lambda\phi}$
- (C) $p = \frac{p^e}{1 + \lambda\phi}$
- (D) $p = \frac{\bar{y}_D - y^{FE}}{1 + \lambda\phi}$

Explanations.

- (A) Wrong: forgot feedback through AD slope ϕ .
- (B) Correct: substitute AD into SRAS and solve for p .
- (C) Wrong: ignores demand term.
- (D) Wrong: ignores p^e .

⑲ Okun numerical. If $u - u^n = -0.5x$ and $x = 2\%$, then $u - u^n$ is B

- (A) $+1\%$
- (B) -1%
- (C) $+4\%$
- (D) -4%

Explanations.

- (A) Wrong sign.
- (B) Correct: $-0.5 \times 2\% = -1\%$.
- (C) Wrong magnitude.

(D) Wrong magnitude.

②0 IS–MP–PC model blocks are B

- (A) IS, LM, and Phillips curve
- (B) IS, monetary policy rule, and Phillips curve
- (C) money supply rule, SRAS, and LRAS
- (D) government budget constraint, money demand, and NKPC

Explanations.

- (A) Wrong: LM is replaced by a policy rule in IS – MP – PC.
 - (B) Correct.
 - (C) Wrong.
 - (D) Wrong.
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Problem 2 (Questions 21–40): IS–MP–PC, Taylor principle, NK

②1 Whelan-style Phillips curve: $\pi_t = \pi_t^e + \gamma(y_t - y_t^*) + \varepsilon_t^\pi$. If π_t^e rises holding gap fixed, then A

- (A) π_t rises one-for-one
- (B) π_t falls one-for-one
- (C) π_t unchanged
- (D) the slope γ changes

Explanations.

- (A) Correct: intercept shifts up.
- (B) Wrong.

- (C) Wrong.
- (D) Wrong: slope unchanged.

②② **Baseline MP rule:** $i_t = r^* + \pi^* + \beta_\pi(\pi_t - \pi^*)$. The Taylor principle typically requires C

- (A) $\beta_\pi < 1$
- (B) $\beta_\pi = 1$
- (C) $\beta_\pi > 1$
- (D) $\beta_\pi < 0$

Explanations.

- (A) Wrong.
- (B) Boundary (real rate doesn't respond to inflation deviations).
- (C) Correct.
- (D) Wrong.

②③ **IS–MP slope sign.** In IS–MP, output falls with inflation (downward IS–MP in (y, π)) when C

- (A) $\beta_\pi < 1$
- (B) $\beta_\pi = 0$
- (C) $\beta_\pi > 1$
- (D) $\gamma = 0$

Explanations.

- (A) Wrong: then IS–MP slopes up (destabilizing).
- (B) Not the key condition.
- (C) Correct: real rate rises when inflation rises.
- (D) Not relevant for IS–MP slope.

②④ Define $\theta = \frac{1}{1 + \alpha\gamma(\beta_\pi - 1)}$ with $\alpha, \gamma > 0$. If $\beta_\pi > 1$, then B

- (A) $\theta > 1$
- (B) $0 < \theta < 1$
- (C) $\theta = 1$
- (D) $\theta < 0$

Explanations.

- (A) Wrong: denominator exceeds 1 so cannot exceed 1.
- (B) Correct.
- (C) Wrong: occurs when $\beta_\pi = 1$ (or $\alpha\gamma = 0$).
- (D) Wrong.

(25) Numerical θ . Let $\alpha = 1, \gamma = 0.5, \beta_\pi = 3$. Then $\theta =$ B

- (A) 0.4
- (B) 0.5
- (C) 2
- (D) -1

Explanations.

- (A) Wrong: used β_π instead of $(\beta_\pi - 1)$.
- (B) Correct: $1/(1 + 0.5 \cdot 2) = 1/2 = 0.5$.
- (C) Wrong.
- (D) Wrong.

(26) Interpretation. A smaller θ means inflation is B

- (A) more sensitive to expectations
- (B) more tightly anchored to the target
- (C) more sensitive to shocks
- (D) independent of policy

Explanations.

- (A) Wrong: smaller θ reduces weight on expectations/shocks.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

(27) Demand shock in IS–MP–PC (one-period). A positive demand shock shifts IS–MP right and typically leads to A

- (A) $y \uparrow$ and $\pi \uparrow$
- (B) $y \downarrow$ and $\pi \uparrow$
- (C) $y \uparrow$ and $\pi \downarrow$
- (D) $y \downarrow$ and $\pi \downarrow$

Explanations.

- (A) Correct.
- (B) Cost-push pattern.
- (C) Wrong.
- (D) Wrong.

(28) Cost-push shock ($\varepsilon^\pi > 0$) typically leads to B

- (A) $y \uparrow$ and $\pi \uparrow$
- (B) $y \downarrow$ and $\pi \uparrow$
- (C) $y \uparrow$ and $\pi \downarrow$
- (D) $y \downarrow$ and $\pi \downarrow$

Explanations.

- (A) Wrong.
- (B) Correct (stagflation pressure).
- (C) Wrong.
- (D) Wrong.

②9 Adaptive expectations assumption is B

- (A) $\pi_t^e = \pi^*$
- (B) $\pi_t^e = \pi_{t-1}$
- (C) $\pi_t^e = \mathbb{E}_t \pi_{t+1}$
- (D) $\pi_t^e = 0$ always

Explanations.

- (A) Wrong (anchored).
- (B) Correct.
- (C) Forward-looking.
- (D) Wrong.

③0 Two-period NK IS equation is A

- (A) $x_0 = \mathbb{E}_0 x_1 - \frac{1}{\sigma}(i_0 - \mathbb{E}_0 \pi_1 - r_0^n)$
- (B) $x_0 = \mathbb{E}_0 x_1 + \frac{1}{\sigma}(i_0 - \mathbb{E}_0 \pi_1 - r_0^n)$
- (C) $x_0 = \frac{1}{\sigma}(i_0 - \mathbb{E}_0 \pi_1 - r_0^n)$
- (D) $x_0 = -\sigma(i_0 - \mathbb{E}_0 \pi_1 - r_0^n)$

Explanations.

- (A) Correct.
- (B) Wrong sign.
- (C) Missing expectation term and wrong sign.
- (D) Wrong scaling.

③1 Two-period NKPC is B

- (A) $\pi_0 = \pi_{-1} + \kappa x_0 + u_0$
- (B) $\pi_0 = \beta \mathbb{E}_0 \pi_1 + \kappa x_0 + u_0$
- (C) $\pi_0 = \kappa x_0 + u_0$ always
- (D) $\pi_0 = \beta \pi_0 + \kappa x_0$

Explanations.

- (A) Backward-looking.
- (B) Correct.
- (C) Only if $\mathbb{E}_0\pi_1 = 0$.
- (D) Wrong.

③② **Calculation (NKPC).** Let $\beta = 0.99$, $\mathbb{E}_0\pi_1 = 0$, $\kappa = 0.5$, $u_0 = 0$, and $x_0 = 2\%$. Then $\pi_0 =$ A

- (A) 1%
- (B) 0.5%
- (C) 2%
- (D) 0%

Explanations.

- (A) Correct: $\pi_0 = 0.5 \times 2\% = 1\%$.
- (B) Wrong: would be $x_0 = 1\%$.
- (C) Wrong: would require $\kappa = 1$.
- (D) Wrong.

③③ **Calculation (Taylor rule).** If $i_0 = \bar{i} + \phi_\pi\pi_0$ with $\bar{i} = 1\%$, $\phi_\pi = 1.5$, and $\pi_0 = 2\%$, then $i_0 =$ C

- (A) 2%
- (B) 3%
- (C) 4%
- (D) 5%

Explanations.

- (A) Wrong: missing reaction term.
- (B) Wrong: used $\phi_\pi = 1$ effectively.
- (C) Correct: $1 + 1.5 \times 2 = 4\%$.

(D) Wrong.

34) Calculation (Fisher). If $i_0 = 4\%$ and $\mathbb{E}_0\pi_1 = 1\%$, then $r_0 \approx$ A

(A) 3%

(B) 5%

(C) -3%

(D) 1%

Explanations.

(A) Correct.

(B) Wrong.

(C) Wrong sign.

(D) Wrong.

35) ZLB. The zero lower bound is the constraint B

(A) $r_0 \geq 0$

(B) $i_0 \geq 0$

(C) $\pi_0 \geq 0$

(D) $x_0 \geq 0$

Explanations.

(A) Wrong: real rates can be negative.

(B) Correct.

(C) Wrong.

(D) Wrong.

36) ZLB implication. If $i_0 = 0$ and expected inflation falls, then the real rate B

(A) falls

(B) rises

(C) is unchanged

(D) becomes negative

Explanations.

- (A) Wrong: $r = i - \mathbb{E}\pi$.
- (B) Correct: lower expected inflation raises r .
- (C) Wrong.
- (D) Wrong: sign depends on inflation expectation.

(37) **Calculation (IS at ZLB).** Suppose $x_1 = 0$, $\sigma = 1$, $r_0^n = 0\%$, $i_0 = 0$, and $\mathbb{E}_0\pi_1 = -1\%$. Then $x_0 =$ C

- (A) 0
- (B) +1%
- (C) -1%
- (D) +2%

Explanations.

- (A) Wrong: real rate gap is positive, so gap is negative.
- (B) Wrong sign.
- (C) Correct: $i - \mathbb{E}\pi - r^n = 0 - (-1) - 0 = 1\%$, so $x_0 = -(1\%) = -1\%$.
- (D) Wrong magnitude.

(38) **Calculation (IS – MP).** If IS is $y = y^* - \alpha(i - \pi - r^*)$ and MP is $i = r^* + \pi^* + \beta_\pi(\pi - \pi^*)$, then the coefficient on $(\pi - \pi^*)$ in IS–MP is A

- (A) $-\alpha(\beta_\pi - 1)$
- (B) $+\alpha(\beta_\pi - 1)$
- (C) $-\alpha\beta_\pi$
- (D) 0

Explanations.

- (A) Correct: substitute MP into IS and simplify $i - \pi - r^* = (\beta_\pi - 1)(\pi - \pi^*)$.
- (B) Wrong sign.
- (C) Wrong: misses the $-\pi$ term when forming the real rate gap.

(D) Wrong: only if $\beta_\pi = 1$.

39) Calculation (policy aggressiveness). If $\alpha = 1$ and $\gamma = 0.5$, what value of β_π makes the boundary $1 - \frac{1}{\alpha\gamma}$ equal to -1 ? C

(A) $\beta_\pi = 0$

(B) $\beta_\pi = 1$

(C) β_π does not enter this boundary; the boundary is -1 given $\alpha\gamma = 0.5$.

(D) $\beta_\pi = 2$

Explanations.

(A) Wrong: boundary depends only on $\alpha\gamma$.

(B) Wrong.

(C) Correct: $1 - 1/(0.5) = -1$ regardless of β_π .

(D) Wrong.

40) Concept. In the NK model, forward-looking behavior is primarily captured by B

(A) $\pi_t = \pi_{t-1} + \kappa x_t$

(B) $x_t = \mathbb{E}_t x_{t+1} - \frac{1}{\sigma}(i_t - \mathbb{E}_t \pi_{t+1} - r_t^n)$ and $\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t$

(C) $M/P = kY - hi$

(D) $y = y^{FE} + \alpha(p - p^e)$

Explanations.

(A) Wrong: backward-looking.

(B) Correct: expectations of next-period variables enter both IS and NKPC.

(C) Wrong: money demand is IS – LM.

(D) Wrong: SRAS is a reduced-form level relation.